Data Structures (CS2001)

Sessional-I Exam

Date: 22nd September, 2026

Total Time (Hrs): 01

Total Marks: 40

Total Questions: 03

Course Instructor(s):

Dr. Anam Qureshi, Dr. Muhammad Yaseen,
Mr. Shoaib Rauf, Mr. Basit Jasani, Ms. Sumaiyah Zahid,
Mr. Noman Rajput, Mr. Sameer Faisal, Mr. Zulfiqar Ali,
Mr. Shafique Rehman, Ms. Zainab Asif Jawed, Mr. Ahmed Qaiser

[redacted] Student Name | [redacted] Roll No | [redacted] Section | [redacted] Student Signature

Do not write below this line

Attempt all the questions in-order.

CLO # 1: *Use & explain* concepts related to basic and advanced data structures and describe their usage in terms of common algorithmic operations

Question 01: **[Estimated Time: 15 minutes] [10 Marks]**

The Global Football Association (GFA) has provided the performance records of teams participating in the FIFA World Cup 2026, containing each team's name and total goals scored. The records are given in random order, for example:

{{"Portugal", 34}, {"Brazil", 52}, {"Argentina", 26}, {"Germany", 29}, {"Spain", 27}, {"France", 34}}

Write a C++ program to implement the following tasks:

- Sort the teams with respect to their goals scored in descending order. [5 Marks]
- If two teams have scored the same number of goals, the team with the lengthier name should appear first. [2 Marks]
- If two teams have scored the same number of goals and their names are of same length, maintain their original order. [2 Marks]
- Display the final sorted list, including each team's name and total number of goals scored order. [1 Mark]

[Note: You may use any sorting algorithm except Bubble Sort and Selection Sort]

CLO # 3: *Compare different data structures in terms of their relative efficiency and design effective solutions and algorithms that make use of them.*

Question 02: **[Estimated Time: 20 minutes] [15 Marks]**

Basheer implemented an adder for 4-digit decimal numbers using a singly linked list. Each number is represented as a singly linked list, where each node stores a single decimal digit (0–9). For example:

L1: 1956 NULL ← 1 ← 9 ← 5 ← 6
L2: 1239 NULL ← 1 ← 2 ← 3 ← 9

L3: 3195 NULL ← 3 ← 1 ← 9 ← 5

Write a C++ program that adds the corresponding digits of L1 and L2 and stores the resulting digits in a third singly linked list, L3, while maintaining their correct positional order.

Use a loop to traverse the nodes, so the same implementation can be easily extended to numbers containing more digits. You need to implement all basic linked-list functionalities required to implement the solution using separate functions. The main() function must demonstrate the complete working of the program.

CLO # 3: *Compare* different data structures in terms of their relative efficiency and *design* effective solutions and algorithms that make use of them.

Question 03: **[Estimated Time: 25 minutes] [15 Marks]**

An examination room contains N students enrolled in three different courses: Data Structures (DS), Computer Organization and Assembly Language (COAL), and Data Science. The students are initially seated in a queue in random order. Each student must submit their answer script according to the FIFO (First-In, First-Out) principle.

As each student submits their answer script, the examiner identifies the student's course and adds the answer script onto the corresponding course-wise stack:

- DS students → DS Stack
- COAL students → COAL Stack
- Data Science students → Data Science Stack

Design and implement a C++ program using linked lists that:

- Implements a Queue to store the students in their initial random order. [1 Mark]
- Implements three separate Stacks, one for each course. [1 Mark]
- Performs the required enqueue(), dequeue(), push(), and pop() operations. [8 Marks]
- Dequeues students from the queue (FIFO) in order. For each dequeued student, identify their course and push their answer script onto the corresponding course-wise stack. [2 Marks]
- After all students have submitted their answer scripts, display the contents of all three course-wise stacks from TOP to BOTTOM. [1 Mark]
- The main() function must demonstrate the complete working of the program by creating the queue, inserting students, processing the queue, and displaying the three resulting stacks. [2 Marks]

========== Best of Luck ==========